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Huitzo × Clarkson University ECE Candidate Research Project Proposals

Vision, Sensors, and Edge AI MS Project Collaboration,
Fall 2026

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Document Purpose

This document presents four candidate research project proposals for joint scoping between Huitzo and the Clarkson University Electrical and Computer Engineering department. The proposals span on-device computer vision for regulated insurance workflows, sovereign facility access control with tamper-evident audit logging, cross-architecture inference benchmarking, and a single-board reference design for a self-hosted edge AI control appliance.

Each proposal is independent and self-contained. Each defines a distinct research contribution, a defensible publication target, and a six-to-eight month execution window aligned with a Fall 2026 MS project cycle. The proposals share a common technical center of gravity in edge inference and data sovereignty, but they do not depend on one another and can be selected individually on their merits.

The goal of this review cycle is to select one proposal to expand into a full project plan with architecture, milestones, and deliverables. The seed concept of camera-based face detection raised in earlier discussion is preserved as the technical core of Proposal 2.

Proposal 1 On-Device Vision Module for Loss Inspection in Property and Casualty Insurance

Abstract

Abstract—Property and Casualty insurance carriers increasingly seek to deploy computer-vision damage classification for adjuster-captured imagery, including roof condition assessment, vehicle exterior damage, and water intrusion documentation. Commercial off-the-shelf solutions typically require uploading imagery to cloud vision APIs, creating compliance friction for carriers subject to data sovereignty regulations and personally identifiable information constraints embedded in claim-site photography. This project proposes the design and evaluation of an edge-deployed vision module that performs damage classification and duplicate or tamper detection entirely within the carrier’s network on a Jetson-class embedded device. The module emits a structured evidence record suitable for ingestion by downstream document-processing workflows. The work will benchmark inference accuracy, end-to-end latency, and power consumption against a cloud-API baseline using publicly available roof-damage and vehicle-damage datasets. The contribution is a reproducible reference design demonstrating measurable sovereignty-preserving vision inference for regulated insurance workflows, accompanied by an empirical comparison to cloud-based alternatives across the dimensions of accuracy, energy use, and end-to-end response time.

Index Terms

Index Terms—edge computing, computer vision, damage classification, data sovereignty, embedded inference, insurance technology, on-device deep learning.

Additional Information

- Hardware: NVIDIA Jetson Orin Nano (8GB) developer kit.
- Frameworks: PyTorch, ONNX, NVIDIA TensorRT, JetPack 7.x.
- Reference datasets: Roboflow roof-damage corpus, KITTI vehicle imagery, comparable publicly available P&C-relevant datasets.
- Timeline: 6–8 months (target completion December 2026).
- Candidate output artifacts: open reference implementation, benchmark report, short demonstration video, optional applied workshop paper submission.

Proposal 2 Sovereign Facility Access Control Using On-Device Face Detection and Tamper-Evident Audit Logging

Abstract

Abstract—Regulated facilities including pharmaceutical cleanrooms, hospital restricted wards, defense suppliers under CMMC 2.0, and SOC 2 controlled environments require auditable physical access logs with biometric verification, yet compliance and contractual constraints frequently prohibit transmitting biometric imagery to third-party cloud services. This project proposes an end-to-end edge system that performs camera-based face detection, badge-presence reconciliation, and dwell-time anomaly detection on a Jetson- or Hailo-class device installed at the facility access point. The system emits cryptographically signed audit log entries using Ed25519 signatures and is architected such that no video frame egresses the facility network. The work will benchmark sovereign edge inference latency, detection accuracy, and energy consumption against equivalent cloud-based baselines. The contribution combines a deployable reference design with a novel signed-audit-trail architecture suitable for regulated physical-access controls. The proposal directly extends the seed concept of camera-based face detection toward a verifiable, compliance-grade, sovereignty-preserving access system with cryptographic auditability.

Index Terms

Index Terms—edge computing, biometric verification, face detection, tamper-evident logging, regulated environments, data sovereignty, physical access control.

Additional Information

- Hardware: NVIDIA Jetson Orin Nano or Hailo-8L M.2 accelerator module; commodity USB or MIPI CSI camera.
- Cryptography: Ed25519 signatures over append-only log entries.
- Candidate model families: BlazeFace, RetinaFace-mobile, YOLO-Face, or comparable lightweight face detectors.
- Timeline: 6–8 months (target completion December 2026).
- Candidate output artifacts: reference implementation, signed-log specification document, latency and accuracy benchmark report, demonstration video, candidate security workshop paper.

Proposal 3 Cross-Architecture Sensor-Input Abstraction and Edge Benchmark for Heterogeneous Inference Runtimes

Abstract

Abstract—Edge artificial-intelligence deployment is increasingly heterogeneous, spanning general-purpose CPUs, integrated GPUs, and dedicated neural accelerators with mutually incompatible software development kits. This fragmentation forces developers to rewrite sensor-driven inference pipelines per target, undermining the portability that motivates edge deployment in the first place. This project proposes a sensor-input abstraction layer that decouples sensor-driven inference modules from their underlying runtime, enabling the same module to execute unchanged across an x86 CPU baseline, an NVIDIA Jetson platform via TensorRT, and a Hailo-8L accelerator via HailoRT. The work will define the abstraction contract, implement a reference vision module that consumes the abstraction, and produce a public benchmark suite reporting end-to-end latency, sustained throughput, power consumption, and inference accuracy across all targets. Results will be prepared for submission to the MLPerf Tiny Open Division and a peer-reviewed systems venue. The contribution is both an open abstraction template and a reproducible benchmark methodology for heterogeneous edge AI runtimes.

Index Terms

Index Terms—edge AI, heterogeneous computing, inference runtime, MLPerf, benchmark methodology, cross-architecture portability, sensor abstraction.

Additional Information

- Hardware: x86 host (laptop or workstation), NVIDIA Jetson Orin Nano, Hailo-8L M.2 module.
- Frameworks: PyTorch, ONNX, NVIDIA TensorRT, HailoRT.
- Candidate publication targets: MLPerf Tiny Open Division (v1.4 or v1.5 cycle), MLSys or EuroSys workshop track, IEEE Edge Computing.
- Timeline: 6–8 months (target completion December 2026).
- Candidate output artifacts: open-source abstraction implementation, benchmark harness, comparative benchmark report, candidate systems paper, MLPerf submission package.

Proposal 4 Single-Board Reference Design for a Self-Hosted AI Control Layer at the Edge

Abstract

Abstract—As organizations subject to data sovereignty constraints adopt artificial intelligence, demand grows for self-hosted, single-appliance deployments of AI control infrastructure capable of operating independently of cloud connectivity. This project proposes the design, implementation, and benchmarking of a single-board reference appliance hosting a complete AI control layer — including an inference engine, a modular runtime, an audit-logging subsystem, and an identity-management component — alongside a representative computer-vision workload. The reference design targets a Jetson AGX Thor or Jetson Orin NX platform, with an optional Hailo-8 vision co-processor for inference offload. The deliverables include a published reference architecture document, a complete bill of materials, and measured benchmarks for power consumption, end-to-end inference latency, and sustained throughput evaluated against equivalent cloud-based baselines. The contribution is a reproducible reference design for a self-hosted edge AI control appliance suitable for regulated and connectivity-limited deployment scenarios.

Index Terms

Index Terms—edge AI, embedded systems, reference design, AI appliance, sovereign computing, system benchmarking, on-premises inference.

Additional Information

- Hardware: NVIDIA Jetson AGX Thor developer kit or NVIDIA Jetson Orin NX 16GB; optional Hailo-8 M.2 co-processor.
- Software: JetPack 7.1, containerized runtime (Docker or containerd), reference vision workload.
- Reference benchmarks: latency, sustained throughput, power consumption, end-to-end response time versus a comparable cloud-API baseline.
- Timeline: 6–8 months (target completion December 2026).
- Candidate output artifacts: reference architecture document, bill of materials, benchmark report, demonstration video, candidate systems or embedded-systems paper.

Next Steps

- Reviewer comments via tracked changes or comments in this document.
- Selection of one proposal for full scoping.
- Follow-up scoping call to define architecture, milestones, and deliverables.